

ERIC SCHMIDT



Contact

Please use the [contact formular](#) to get in touch.

About Me

I am a dedicated and results-driven Developer, with 5 years of experience in research. My passion for programming and prototyping has led me to develop a strong foundation in Computer Vision and Robotics, which I have successfully applied in several projects.

I thrive in dynamic environments and am committed to delivering high-quality work, whether working independently or as part of a team.

EDUCATION

10/2020 - 09/2022

M.Sc. • Nanobiophysics

Technical University of Dresden

Master thesis:

Development of an AI-based Python software for the automatic segmentation of 4D microscopy time series with automated graph-based error correction and subsequent quantification and visualization.

10/ 2017 - 09/2020

B.Sc. • Physics

Technical University of Dresden

Bachelor thesis:

Software Development (MATLAB) for the analysis of tracking data, to quantify and visualize single-cell movement.

TECHNICAL SKILLS

GitHub/GitLab	daily use
Python - scientific*	4.5 years
Python - AI stack**	3.5 years
ARDUINO (C++)	2.5 years
Microscopy	2.5 years
MATLAB	1.5 years
RASPBERRY PI	1.5 year
SQL	1 year
Blender (+Python)	1 year
D3.js/javascript	1/2 year
HTML, CSS	1/2 year
Cloud	1/2 year
R	1/2 year
LabView	1/2 year
AWORK/MIRO	1/2 year
KNIME, PySPARK	3 months
MS Office, LaTeX	many years

CLICK ME



WORK EXPERIENCE

01/2022 - 06/2023

Data Scientist • Clade (Esslingen, DE)

- Data processing with chemometric techniques
- Fixing of bugs and problems in existing algorithms
- Implementation of AI signal denoising
- Python, R, Databases

11/2022 - 11/2023

PhD • EPFL (Lausanne, CH)

- Maintaining (hardware) and improving (software) the advanced microscopy setup (interferometric scattering microscopy)
- Bacteria cultivation (S2) and sample preparation, as well as imaging
- Image segmentation and particle tracking via Python and CNNs
- Data Quantification
- Database management (SQL)
- Code and Python Package development for Nature Microbiology publication [1]

LANGUAGES

German	Native
English	C1
Russian	A2
Spanish	A2

PUBLIC APPEARANCE

- Scholarship (Deutschlandstipendium)
- Best student in the international master course [2]
- FameLab Dresden [3]
- Production of teaching videos for programming beginners and scientists (YouTube [4], GitHub [5])

[1] <https://doi.org/10.1101/2024.09.09.611974>

[2] <https://youtu.be/CuNeU-6LI6E?t=44>

[3] https://www.youtube.com/watch?v=b_WJ6wfkoozU&t=1392s

[4] <https://www.youtube.com/watch?v=LrYPpIok1v8&t=26s>

[5] <https://github.com/Nablaaa/Lensless-Microscope>

*numpy, scipy, matplotlib/seaborn/plotly, pandas/polars, scikit-image, opencv, napari, pytest

** keras, tensorflow, pytorch, scikit-learn

SOFTWARE PROJECTS

12/2021 - 11/2022

Student Job • MPI-CBG (Dresden)

- Develop modular software for the analysis of 3D fluorescent microscopy data software
- CNN based image segmentation, automatic error detection and correction, object tracking, quantification and visualization
- Python, Streamlit, Napari

10/2021 - 01/2022

Internship • B Cube (Dresden)

- Writing Software for object detection and intensity quantification in images
- Using Decision Trees and NNs for molecule-protein binding affinity prediction and drug recommendation
- Python, KNIME

09/2021 - 11/2021

Internship • CSBD (Dresden)

- Testing and optimization of classical and AI based segmentation algorithms for 3D fluorescence microscopy images
- Python

04/2019 - 10/2019

Student Job • IAPP (Dresden)

- Writing Software for calculating the angle dependent quantum efficiency of organic LEDs (OLEDs) that was used to optimize the production process of OLEDs in smart devices
- Python

TEAM PROJECT

03/2022 - 10/2022

international Genetically Engineered Machine (iGEM) [6, 7]

- Developing of a hydrogel based wound dressing that releases growth factors to support the healing of chronic wounds
- Software Development for hydrogel mesh size estimations
- Diffusion simulations
- Team Leading
- Python, COMSOL, LAMMPS

SOFTWARE + HARDWARE

03/2021 - 04/2021

Internship • Max Bergmann Center of Biomaterials (Dresden)

- Development of Hydrogel and Enzyme based biosensors
- Software Development for Sensor Readout and concentration quantification via cyclic voltammetry
- Python, Arduino (C++), Biosensor

02/2020 - 03/2021

Student Job • B Cube (Dresden)

- Cultivation of single cell organisms (Diatoms)
- Imaging and tracking of Diatoms
- Software development for the analysis of the tracking data
- Matlab, Microscopy (TIRF)

01/2021 - 05/2021

Internship • Biotec (Dresden)

- Micro-controller programming for lab specific tasks (sensor readout, actor control, communication)
- Development of a web controlled robot arm that performs titration measurements and quantifies the results on the fly
- Arduino, Raspberry Pi, Python, Robotics

HOME PROJECTS

- Web controlled plant watering system with database for sensor measurements and data driven watering recommendations (Python, RaspberryPi)
- Development of a programming course that teaches how to build and control a cheap, digital, lensless microscope (Python, YouTube, 3D Printing)
- Development of software for drones to detect and follow objects (Python, DJI Tello)

[6]

<https://video.igem.org/w/bAauQTU55NAH849cRH61uS>

[7] <https://2022.igem.wiki/tu-dresden/>